

Immanuel Trummer

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Education

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| École Polytechnique Fédérale de Lausanne (EPFL)
PhD in Computer Science; Advisor: Christoph Koch.
<i>EPFL Thesis Special Distinction.</i> | 2010-2016 |
| University of Stuttgart & École Centrale de Nantes
Double Diploma in Computer Science and Engineering.
<i>Graduated with Distinction.</i> | 2003-2010 |

Academic Appointments

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| Cornell University
Associate Professor, Cornell Bowers CIS. | 2024-Present |
| Cornell University
Assistant Professor, Cornell Bowers CIS. | 2016-2024 |
| École Polytechnique Fédérale de Lausanne (EPFL)
Graduate Research Assistant, LIA/DATA Lab. | 2010-2016 |

Publications

Books

- Immanuel Trummer. [Data Analysis with LLMs: Text, Tables, Images and Sound](#). Manning Publications, 2025, pp. 1–232.

Refereed Conference and Journal Publications¹

- Manuel Schoenberger, Immanuel Trummer, and Wolfgang Mauerer. [Large-Scale Multiple Query Optimisation with Incremental Quantum\(-Inspired\) Annealing](#). In: *SIGMOD* 3 (4 2026), pp. 1–25.
- Jiale Lao and Immanuel Trummer. [SQLBarber: A System Leveraging Large Language Models to Generate Customized and Realistic SQL Workloads](#). In: *SIGMOD* 4 (1 2026), pp. 1–27.
- Manuel Schönberger, Immanuel Trummer, and Wolfgang Mauerer. [Hybrid Mixed Integer Linear Programming for Large-Scale Join Order Optimisation](#). In: *PVLDB* 19 (3 2026), pp. 348–360.
- Jiale Lao, Andreas Zimmerer, Olga Ovcharenko, Tianji Cong, Matthew Russo, Gerardo Vitagliano, Michael Cochez, Fatma Özcan, Gautam Gupta, Thibaud Hottelier, H. V. Jagadish, Kris Kissel, Sebastian Schelter, Andreas Kipf, and Immanuel Trummer. [SemBench: A Benchmark for Semantic Query Processing Engines](#). In: *PVLDB* 19 (8 2026), pp. 1754–1767.
- Immanuel Trummer. [Optimal Block Nested Loops Implementations for Semantic Joins](#). In: *Data Engineering Bulletin* 50 (1 2026), pp. 74–87.
- Immanuel Trummer. [ThalamusDB: Semantic Approximate Query Processing](#). In: *Data Engineering Bulletin* 50 (2 2026), pp. 61–74.
- Immanuel Trummer. [Generating Highly Customizable Python Code for Data Processing with Large Language Models](#). In: *VLDBJ* 34 (21 2025), pp. 1–19.
- Victor Giannakouris and Immanuel Trummer. [Lambda-Tune: Harnessing Large Language Models for Automated Database System Tuning](#). In: *SIGMOD* 3 (1 2025), pp. 1–26.

¹PVLDB publications are presented yearly at the VLDB conference.

- Saehan Jo and Immanuel Trummer. [SpareLLM: Automatically Selecting Task-Specific Minimum-Cost Large Language Models under Equivalence Constraint](#). In: *SIGMOD* 3 (3 2025), pp. 1–26.
- Immanuel Trummer. [Customizing Operator Implementations for SQL Processing via Large Language Models](#). In: *Data Engineering Bulletin* 49 (1 2025), pp. 45–56.
- Tharushi Jayasekara and Immanuel Trummer. [Demonstrating CEDAR: A System for Cost-Efficient Data-Driven Claim Verification](#). In: *SIGMOD*. 2025, pp. 1–4.
- Jiale Lao and Immanuel Trummer. [Demonstrating SQLBarber: Leveraging Large Language Models to Generate Customized and Realistic SQL Workloads](#). In: *SIGMOD*. 2025, pp. 151–154.
- Victor Giannakouris and Immanuel Trummer. [SwellDB: Dynamic Query-Driven Table Generation with Large Language Models](#). In: *SIGMOD*. 2025, pp. 95–98.
- Immanuel Trummer. [DB-BERT: Making Database Tuning Tools “Read” the Manual](#). In: *VLDB Journal* 33 (2024), pp. 1085–1104.
- Saehan Jo and Immanuel Trummer. [ThalamusDB: Approximate Query Processing on Multi-Modal Data](#). In: *SIGMOD* 2 (3 2024), pp. 1–26.
- Ziyun Wei and Immanuel Trummer. [ROME: Robust Query Optimization via Parallel Multi-Plan Execution](#). In: *SIGMOD* 2 (3 2024), pp. 1–25.
- Junxiong Wang, Kaiwen Wang, Yueying Li, Nathan Kallus, Immanuel Trummer, and Wen Sun. [JoinGym: An Efficient Join Order Selection Environment](#). In: *RLJ* 1 (2024), pp. 64–91.
- Victor Giannakouris and Immanuel Trummer. [DBG-PT: A Large Language Model Assisted Query Performance Regression Debugger](#). In: *PVLDB* 17 (12 2024), pp. 4337–4340.
- Immanuel Trummer. [Generating Succinct Descriptions of Database Schemata for Cost-Efficient Prompting of Large Language Models](#). In: *PVLDB* 17 (11 2024), pp. 3511–3523.
- Immanuel Trummer. [Large Language Models: Principles and Practice](#). In: *ICDE*. 2024, pp. 5354–5357.
- Victor Giannakouris and Immanuel Trummer. [Demonstrating -Tune: Exploiting Large Language Models for Workload-Adaptive Database System Tuning](#). In: *SIGMOD*. 2024, pp. 508–511.
- Junxiong Wang, Immanuel Trummer, Ahmet Kara, and Dan Olteanu. [ADOPT: Adaptively Optimizing Attribute Orders for Worst-Case Optimal Join Algorithms via Reinforcement Learning](#). In: *PVLDB* 16 (11 2023), pp. 2805–2817.
- Immanuel Trummer. [Demonstrating GPT-DB: Generating Query-Specific and Customizable Code for SQL Processing with GPT-4](#). In: *PVLDB* 16 (12 2023), pp. 4098–4101.
- Junxiong Wang, Immanuel Trummer, Ahmet Kara, and Dan Olteanu. [Demonstrating ADOPT: Adaptively Optimizing Attribute Orders for Worst-Case Optimal Joins via Reinforcement Learning](#). In: *PVLDB* 16 (12 2023), pp. 4094–4097.
- Simran Arora, Brandon Yang, Sabri Eyuboglu, Avanika Narayan, Andrew Hojel, Immanuel Trummer, and Christopher Re. [Language Models Enable Simple Systems for Generating Structured Views of Heterogeneous Data Lakes](#). In: *PVLDB* 17 (2 2023), pp. 92–105.
- Immanuel Trummer. [Can Large Language Models Predict Data Correlations from Column Names?](#) In: *PVLDB* 16 (13 2023), pp. 4310–4323.
- Manuel Schoenberger, Immanuel Trummer, and Wolfgang Mauerer. [Quantum-Inspired Digital Annealing for Join Ordering](#). In: *PVLDB* 17 (3 2023), pp. 511–524.
- Immanuel Trummer. [Demonstrating NaturalMiner: Searching Large Data Sets for Abstract Patterns Described in Natural Language](#). In: *SIGMOD*. 2023, pp. 139–142.
- Immanuel Trummer. [BABOONS: Black-Box Optimization of Data Summaries in Natural Language](#). In: *PVLDB* 15 (11 2022), pp. 2980–2993.
- Ziyun Wei and Immanuel Trummer. [SkinnerMT: Parallelizing for Efficiency and Robustness in Adaptive Query Processing on Multicore Platforms](#). In: *PVLDB* 16 (4 2022), pp. 905–917.
- Immanuel Trummer. [CodexDB: Synthesizing Code for Query Processing from Natural Language Instructions using GPT-3 Codex](#). In: *PVLDB* 15 (11 2022), pp. 2921–2928.
- Immanuel Trummer. [From BERT to GPT-3 Codex: Harnessing the Potential of Very Large Language Models for Data Management](#). In: *PVLDB* 15 (12 2022), pp. 3770–3773.
- Junxiong Wang, Debabrota Basu, and Immanuel Trummer. [Procrastinated Tree Search: Black-box Optimization with Delayed, Noisy, and Multi-fidelity Feedback](#). In: *AAAI* 36 (9 2022), pp. 10381–10390.
- Immanuel Trummer. [Towards NLP-Enhanced Data Profiling Tools](#). In: *CIDR*. 2022, pp. 1–1.

- Immanuel Trummer. [DB-BERT: a Database Tuning Tool that “Reads the Manual”](#). In: *SIGMOD*. 2022, pp. 190–203.
- Immanuel Trummer. [Demonstrating DB-BERT: A Database Tuning Tool that “Reads” the Manual](#). In: *SIGMOD*. 2022, pp. 2437–2440.
- Immanuel Trummer. [Database Tuning using Natural Language Processing](#). In: *SIGMOD Record* 50 (3 2021), pp. 27–28.
- Junxiong Wang, Immanuel Trummer, and Debabrota Basu. [UDO: Universal Database Optimization using Reinforcement Learning](#). In: *PVLDB* 14 (13 2021), pp. 3402–3414.
- Ziyun Wei, Immanuel Trummer, and Connor Anderson. [Robust Voice Querying with MUVE: Optimally Visualizing Results of Phonetically Similar Queries](#). In: *PVLDB* 14 (11 2021), pp. 2397–2409.
- Immanuel Trummer. [The Case for NLP-Enhanced Database Tuning: Towards Tuning Tools that “Read the Manual”](#). In: *PVLDB* 14 (7 2021), pp. 1159–1165.
- Immanuel Trummer. [WebChecker: Towards an Infrastructure for Efficient Misinformation Detection at Web Scale](#). In: *IEEE Data Eng. Bull.* 44 (3 2021), pp. 66–77.
- Immanuel Trummer, Junxiong Wang, Ziyun Wei, Deepak Maram, Samuel Moseley, Saehan Jo, Joseph Antonakakis, and Ankush Rayabhari. [SkinnerDB: Regret-bounded Query Evaluation via Reinforcement Learning](#). In: *ACM Transactions on Database Systems* 46 (3 2021), pp. 1–45.
- Immanuel Trummer and Connor Anderson. [Optimally Summarizing Data by Small Fact Sets for Concise Answers to Voice Queries](#). In: *ICDE*. 2021, pp. 1715–1726.
- Junxiong Wang, Immanuel Trummer, and Debabrota Basu. [Demonstrating UDO: A Unified Approach for Optimizing Transaction Code, Physical Design, and System Parameters via Reinforcement Learning](#). In: *SIGMOD*. 2021, pp. 2794–2797.
- Ziyun Wei, Immanuel Trummer, and Connor Anderson. [Demonstrating Robust Voice Querying with MUVE: Optimally Visualizing Results of Phonetically Similar Queries](#). In: *SIGMOD*. 2021, pp. 2798–2802.
- Georgios Karagiannis, Mohammed Saeed, Paolo Papotti, and Immanuel Trummer. [Scrutinizer: Fact Checking Statistical Claims](#). In: *PVLDB* 13 (12 2020), pp. 2965–2968.
- Georgios Karagiannis, Immanuel Trummer, Saehan Jo, Shubham Khandelwal, Xuezhi Wang, and Cong Yu. [Mining an “Anti-Knowledge Base” from Wikipedia Updates with Applications to Fact Checking and Beyond](#). In: *PVLDB* 13 (4 2020), pp. 561–573.
- Immanuel Trummer. [Demonstrating the Voice-Based Exploration of Large Data Sets with CiceroDB-Zero](#). In: *PVLDB* 13 (12 2020), pp. 2869–2872.
- Georgios Karagiannis, Mohammed Saeed, Paolo Papotti, and Immanuel Trummer. [Scrutinizer: A Mixed-Initiative Approach to Large-Scale, Data-Driven Claim Verification](#). In: *PVLDB* 13 (12 2020), pp. 2508–2521.
- Saehan Jo, Jialing Pei, and Immanuel Trummer. [Demonstration of ScroogeDB: Getting More Bang for the Buck with Deterministic Approximation in the Cloud](#). In: *PVLDB* 13 (12 2020), pp. 2961–2964.
- Georgios Karagiannis, Mohammed Saeed, Paolo Papotti, and Immanuel Trummer. [Scrutinizer: a System for Checking Statistical Claims](#). In: *BDA*. 2020, pp. 72–73.
- Saehan Jo and Immanuel Trummer. [BitGourmet: Deterministic Approximation via Optimized Bit Selections](#). In: *CIDR*. 2020, pp. 1–7.
- Saehan Jo and Immanuel Trummer. [Demonstration of BitGourmet: Data Analysis via Deterministic Approximation](#). In: *SIGMOD*. 2020, pp. 2801–2804.
- Saehan Jo, Immanuel Trummer, Weicheng Yu, Xuezhi Wang, Cong Yu, Daniel Liu, and Niyati Mehta. [AggChecker: a Fact-Checking System for Text Summaries of Relational Data Sets](#). In: *PVLDB* 12 (12 2019), pp. 1938–1941.
- Immanuel Trummer. [Data Vocalization with CiceroDB](#). In: *CIDR*. 2019, pp. 1–8.
- Saehan Jo, Immanuel Trummer, Weicheng Yu, Xuezhi Wang, Cong Yu, Daniel Liu, and Niyati Mehta. [Verifying Text Summaries of Relational Data Sets](#). In: *SIGMOD*. 2019, pp. 299–316.
- Immanuel Trummer, Yicheng Wang, and Saketh Mahankali. [A Holistic Approach for Query Evaluation and Result Vocalization in Voice-Based OLAP](#). In: *SIGMOD*. 2019, pp. 936–953.
- Immanuel Trummer, Junxiong Wang, Deepak Maram, Samuel Moseley, Saehan Jo, and Joseph Antonakakis. [SkinnerDB: Regret-Bounded Query Evaluation via Reinforcement Learning](#). In: *SIGMOD*. 2019, pp. 1039–1050.
- Immanuel Trummer. [Exact Cardinality Query Optimization with Bounded Execution Cost](#). In: *SIGMOD*. 2019, pp. 2–17.

- Immanuel Trummer, Samuel J Mosley, Joseph Antonakakis, and Saehan Jo. [SkinnerDB: Regret-Bounded Query Evaluation via Reinforcement Learning](#). In: *PVLDB* 11 (12 2018), pp. 2074–2077.
- Immanuel Trummer, Mark Bryan, and Ramya Narasimha. [Vocalizing Large Time Series Efficiently](#). In: *PVLDB* 11 (11 2018), pp. 1563–1575.
- Immanuel Trummer and Christoph Koch. [Multi-Objective Parametric Query Optimization](#). In: *Vldb Journal* 26 (1 2017), pp. 107–124.
- Immanuel Trummer, Jiancheng Zhu, and Mark Bryan. [Data Vocalization: Optimizing Voice Output of Relational Data](#). In: *PVLDB* 10 (11 2017), pp. 1574–1585.
- Immanuel Trummer and Christoph Koch. [Multi-Objective Parametric Query Optimization](#). In: *Communications of the ACM* 60 (10 2017), pp. 81–89.
- Immanuel Trummer and Christoph Koch. [Solving the Join Ordering Problem via Mixed Integer Linear Programming](#). In: *SIGMOD*. 2017, pp. 1025–1040.
- Immanuel Trummer and Christoph Koch. [Parallelizing Query Optimization on Shared-Nothing Architectures](#). In: *PVLDB* 9 (9 2016), pp. 660–671.
- Immanuel Trummer and Christoph Koch. [Multiple Query Optimization on the D-Wave 2X Adiabatic Quantum Computer](#). In: *PVLDB* 9 (9 2016), pp. 648–659.
- Immanuel Trummer and Christoph Koch. [Multi-Objective Parametric Query Optimization](#). In: *ACM SIGMOD Research Highlights* 45 (1 2016), pp. 24–31.
- Immanuel Trummer and Christoph Koch. [A Fast Randomized Algorithm for Multi-Objective Query Optimization](#). In: *SIGMOD*. 2016, pp. 1737–1752.
- Immanuel Trummer, Alon Halevy, Hongrae Lee, Sunita Sarawagi, and Rahul Gupta. [Mining Subjective Properties on the Web](#). In: *SIGMOD*. 2015, pp. 1745–1760.
- Immanuel Trummer and Christoph Koch. [An Incremental Anytime Algorithm for Multi-Objective Query Optimization](#). In: *SIGMOD*. 2015, pp. 1941–1953.
- Immanuel Trummer and Christoph Koch. [Multi-Objective Parametric Query Optimization](#). In: *PVLDB* 8 (3 2014), pp. 221–232.
- Immanuel Trummer and Christoph Koch. [Approximation Schemes for Many-Objective Query Optimization](#). In: *SIGMOD*. 2014, pp. 1299–1310.
- Immanuel Trummer, Boi Faltings, and Walter Binder. [Multi-Objective Quality-Driven Service Selection - A Fully Polynomial Time Approximation Scheme](#). In: *IEEE Transactions on Software Engineering* 40 (2 2013), pp. 167–191.
- Mehdi Riahi, Thanasis Papaioannou, Immanuel Trummer, and Karl Aberer. [Utility-Driven Data Acquisition in Participatory Sensing](#). In: *EDBT*. 2013, pp. 251–262.
- Immanuel Trummer and Boi Faltings. [Dynamically Selecting Composition Algorithms for Economical Composition as a Service](#). In: *ICSOC*. 2011, pp. 513–522.
- Immanuel Trummer and Boi Faltings. [Optimizing the Tradeoff between Discovery, Composition, and Execution Costs in Service Composition](#). In: *ICWS*. 2011, pp. 476–483.
- Walter Binder, Daniele Bonetta, Cesare Pautasso, Achille Peternier, Diego Milano, Heiko Schuldt, Nenad Stojnic, Boi Faltings, and Immanuel Trummer. [Towards Self-Organizing Service-Oriented Architectures](#). In: *IEEE World Congress on Services*. 2011, pp. 115–121.
- Immanuel Trummer, Frank Leymann, Ralph Mietzner, and Walter Binder. [Cost-Optimal Outsourcing of Applications into the Clouds](#). In: *CloudCom*. 2010, pp. 135–142.

Refereed Workshop Publications

- Victor Giannakouris and Immanuel Trummer. [Breaking Database Lock-in: Agentic Regeneration of High-Performance Storage Readers for Database Bypass](#). In: *AIDB*. 2026, pp. 1–8.
- Immanuel Trummer. [Leveraging Quantum Annealing for Materialized View Selection](#). In: *Q-Data*. 2026, pp. 3–8.
- Immanuel Trummer. [Quantum Annealing for Multiobjective Query Optimization](#). In: *Q-Data*. 2026, pp. 58–58.
- Immanuel Trummer. [Towards Caching Strategies for Quantum Simulators](#). In: *QCDKM*. 2026, pp. 1–2.
- Victor Giannakouris and Immanuel Trummer. [Rethinking Pluggable Federated Query Optimization: From Laptops to Data Warehouse](#). In: *CDMS*. 2025.

- Immanuel Trummer. [Leveraging Quantum Computing for Optimal Data Allocation in Distributed Systems](#). In: *Q-Data*. ACM, 2025, pp. 10–17.
- Immanuel Trummer. [Cost-Based Query Optimization for Quantum Computation](#). In: *Q-Data*. ACM, 2025, pp. 1–2.
- Ibrahim Sabek and Immanuel Trummer. [Second Workshop on Quantum Computing and Quantum-Inspired Technology for Data-Intensive Systems and Applications \(Q-Data\)](#). In: *SIGMOD Companion*. 2025, pp. 882–883.
- Immanuel Trummer and Davide Venturelli. [Leveraging Quantum Computing for Database Index Selection](#). In: *Q-Data*. 2024, pp. 14–26.
- Immanuel Trummer. [Towards Out-of-Core Simulators for Quantum Computing](#). In: *Q-Data*. 2024, pp. 1–1.
- Ibrahim Sabek, Immanuel Trummer, and Stefan Prestel. [First Workshop on Quantum Computing and Quantum-Inspired Technology for Data-Intensive Systems and Applications \(Q-Data\)](#). In: *SIGMOD*. 2024, pp. 663–664.
- Manuel Schoenberger, Immanuel Trummer, and Wolfgang Maurer. [Quantum Optimisation of General Join Trees](#). In: *QDSM*. 2023, pp. 1–12.
- Victor Giannakouris and Immanuel Trummer. [Building Learned Federated Query Optimizers](#). In: *VLDB PhD Workshop*. 2022, pp. 1–5.

Patents

- Immanuel Trummer. [Verifying Text Summaries of Relational Data Sets \(US Patent Number 11113275\)](#). 2021.
- Davide Venturelli and Immanuel Trummer. [System and Method to Hardcode Integer Linear Optimization Problems on Physical Implementations of the Ising Model \(US Patent Number 10691771\)](#). 2020.

Theses

- Immanuel Trummer. [From Massive Parallelization to Quantum Computing: Seven Novel Approaches to Query Optimization](#). PhD thesis. EPFL, 2016, pp. 1–253.
- Immanuel Trummer. [Cost-Optimal Provisioning of Cloud Applications](#). PhD thesis. University of Stuttgart, 2010, pp. 1–98.

Awards and Honors

- Best of VLDB Paper, 2024.
- SIGMOD Best Demo Runner Up Award, 2023.
- NSF CAREER Award, 2022.
- BDA Best Demo Award, 2020.
- Best of SIGMOD Paper (TODS), 2019.
- Google Faculty Research Award, 2018.
- Jim Gray Award, Honorable Mention, 2017.
- EPFL PhD Thesis Special Distinction, 2017.
- Google Faculty Research Award, 2016.
- ACM SIGMOD Research Highlight, 2015.
- Best of VLDB Paper, 2015.
- Google PhD Fellowship, 2015.
- EPFL Teaching Assistant Award, 2015.
- First Graduation Prize, 2010.
- German National Academic Foundation, 2008.
- Scholarship for Academic Excellence, 2007.
- Scholarship for TIME Double Degree, 2005.

Funded Proposals

Internal

- CCSS Grant: “Towards a National Infrastructure of Federated Private Computation Spaces” (2026-2027). PI: Sara Emery. Co-PIs: Steven Wolf, Renata Ivanek, Immanuel Trummer. Total Amount: \$60,000.
- CIDA RIF Seed Grant: “Co-Creating a Human-Machine Interface Better Adapted for On-Farm Data Recording, Curation, Management, and Use” (2021-2023). PI: Louis Longchamp. Co-PIs: Wendy Ju, Immanuel Trummer, Diane Bailey, Mike Stanyard, Erik Smith, Michael Hunter. Total Amount: \$150,000.
- Huawei Research Initiative: “Deterministic Approximation via BitGourmet” (2019). PI: Immanuel Trummer. Total Amount: \$116,831.

External

- FDA Grant: “SAGE: An Integrated Monitoring System for Antimicrobial Use and Stewardship” (2027-2031). PI: Renata Ivanek Miojevic. Co-PIs: Daryl Nydam, Randall Singer, Casey Cazer, Jennifer Granick, Steven A. Wolf, Kurt B. Waldman, Danah Michele Boyd, Daniel Susser, Hakim Weatherspoon, Christopher Matthew De Sa, Immanuel Trummer, Jennifer J. Sun. Total Amount: \$997,668.
- [LinkedIn Grant: “Making Agentic Data Analysis Token-Efficient”](#) (2026-2027). PI: Immanuel Trummer. Total Amount: \$100,000.
- [NSF 2239326: “CAREER: Mining Hints from Text Documents to Guide Automated Database Performance Tuning”](#) (2023-2028). PI: Immanuel Trummer. Total Amount: \$594,874.
- [NSF 1910830: “III: Small: Regret-Bounded Query Evaluation via Reinforcement Learning”](#) (2019-2022). PI: Immanuel Trummer. Total Amount: \$499,999.
- Google AI-Powered Immediate Response to Pandemics Initiative: “Data-driven Automatic Fact Checking of Coronavirus Claims” (2020). PIs: Immanuel Trummer, Paolo Papotti (Co-PI). Total Amount: \$120,000.
- Fact-Checking Development Grant (IFCN & YouTube): “Automatic Fact-Checking of Coronavirus Claims” (2020). PIs: Paolo Papotti, Immanuel Trummer (Co-PI). Total Amount: \$49,800.
- [Google Faculty Research Award: “Mining an ‘Anti-Knowledge Base’ for Fact Checking from Wikipedia Updates”](#) (2018). PI: Immanuel Trummer. Total Amount: \$65,235.
- [Google Faculty Research Award: “Optimizing Voice-Based Output of Relational Data”](#) (2016). PI: Immanuel Trummer. Total Amount: \$57,407.

Talks

Tutorials

- 5/14/24: [Tutorial on Large Language Models: Principles and Practice](#) (90 Minutes), ICDE 2024.
- 3/8/23: [From BERT to GPT-3 Codex: Harnessing the Potential of Very Large Language Models for Data Management](#) (180 Minutes), BTW 2023.
- 9/8/22: [From BERT to GPT-3 Codex: Harnessing the Potential of Very Large Language Models for Data Management](#) (90 Minutes), VLDB 2022.

Panels

- 9/14/2026: [{Quantum, Databases} for {Databases, Quantum}](#), QCE 2026.
- 9/3/2026: [Data Agents: Rethinking Data Systems in the AI Agent Era](#), VLDB 2026.
- 6/5/2026: [10th year DEEM: Looking Back, and Looking Forward](#), DEEM 2026.
- 6/2/2026: [From Queries to Qubits: Data Management Meets Quantum Computing](#), SIGMOD 2026.
- 6/1/2026: [New Researcher Symposium](#), SIGMOD 2026.

- 6/24/2025: [AI for Future Databases: A New Beginning or a Boulevard of Broken Dreams?](#) SIGMOD 2025.
- 6/14/2024: Foundation Models for and in Databases, SIGMOD 2024 Workshop on Exploiting Artificial Intelligence Techniques for Data Management.
- 5/13/2024: [Knowledge Graphs and LLMs: A Relationship under Investigation](#), ICDE 2024 Workshop on Search, Exploration, and Analysis in Heterogeneous Datastores Graph Edition.
- 6/18/23: [Data Management Challenges in LLM-Powered Solutions](#), SIGMOD 2023 Workshop on Data Management for End-to-End Machine Learning.
- 4/11/22: [Futuristic Data Interfaces - Voice Interfaces for Data Access](#), Conference on Database Systems for Advanced Applications 2022.
- 10/3/19: [Anticipating Societal Impact](#), Cornell CIS Celebration 2019.

Invited Talks

- 9/4/2026: [Solving Hard Database Optimization Problems via Digital and Quantum Annealing](#), **Keynote** at QC&DKM.
- 9/4/2026: [Towards AI-Generated Database Management Systems](#), **Keynote** at CDMS.
- 8/31/2026: [Towards Efficient Semantic Query Processing Engines](#), **Keynote** at AIDB.
- 6/5/2026: [Making Making LLM-Based User Interfaces Cost-Efficient](#), **Keynote** at HILDA.
- 6/5/2026: [SemBench: A Benchmark for Semantic Query Processing Engines](#), DEEM.
- 11/5/25: CheaPT: Using Language Models Without Breaking the Bank, University of California, Berkeley.
- 10/30/25: CheaPT: Using Language Models Without Breaking the Bank, Arizona State University (Virtual).
- 9/18/25: SemBench: A Benchmark for Semantic Query Processing Engines, Google (Virtual).
- 7/7/25: CheaPT: Using Language Models Without Breaking the Bank, Snowflake ML Lunch (Virtual).
- 7/2/25: CheaPT: Using Language Models Without Breaking the Bank, Joint Database Systems Seminar by TU Nuremberg, Hasso-Plattner Institute, and TU Darmstadt (Virtual).
- 3/4/25: [CheaPT: Using Language Models Without Breaking the Bank](#), Invited talk at ML4Sys and Sys4ML.
- 11/12/24: [Novel Applications for Large Language Models in Data Management](#), **Keynote** at KBC-LM.
- 6/14/24: [Novel Applications for Large Language Models in Data Management](#), **Keynote** at aiDM.
- 5/23/24: [Novel Applications for Large Language Models in Data Management](#), **Keynote** at NEDB.
- 4/4/24: Novel Use Cases for Large Language Models in Data Management, UIUC (Virtual).
- 12/15/23: [Next-Generation Data Management with Large Language Models](#), Invited talk at TRL.
- 9/1/23: [Applications for Large Language Models in Data Management](#), **Keynote** at AIDB.
- 9/1/23: [Playing Dice with the Universe: a Brief History of Quantum Computing and Potential Use Cases in Data Management](#), **Keynote** at QDSM.
- 7/14/23: Novel Use Cases for Large Language Models in Data Management, TUM.
- 7/13/23: Novel Use Cases for Large Language Models in Data Management, MPI.
- 7/11/23: Towards Highly Customizable Database Systems via Generative AI, University of Stuttgart.
- 7/7/23: Novel Use Cases for Large Language Models in Data Management, TU Berlin.
- 7/5/23: [Next-Generation Data Management via Large Language Models](#), **Keynote** at ScaDS.AI.
- 4/18/23: Towards Highly Customizable Database Systems via Generative AI, Boston University.
- 4/19/23: [Towards Highly Customizable Database Systems via Generative AI](#), MIT.
- 4/18/23: Towards Highly Customizable Database Systems via Generative AI, Harvard.
- 4/7/23: Towards Tuning Tools that “Read” the Manual, University of Buffalo.
- 4/3/23: [Towards AI-Generated Database Management Systems](#), **Keynote** at DBML.
- 3/7/23: Towards AI-Generated Database Management Systems, BTW 2023 Tutorial Workshop on ML for Systems and Systems for ML.
- 2/25/23: Leveraging Generative AI for Data Processing, CWI (Virtual).
- 12/14/22: Towards Tuning Tools that “Read” the Manual, TU Darmstadt (Virtual).
- 12/1/22: Towards Tuning Tools that “Read” the Manual, NYU.
- 11/3/22: Towards Tuning Tools that “Read” the Manual, OSU (Virtual).
- 8/15/22: Novel Applications for Language Models in Data Management, Stanford DB Group.

- 8/3/22: Intra-Query Learning for Worst-Case Optimal Join Algorithms, FDB Workshop (Virtual).
- 3/11/22: Mining Hints for Database Performance Tuning from the Web, Aarhus University (Virtual).
- 7/21/22: Towards Tuning Tools that "Read" the Manual, RelationalAI (Virtual).
- 11/17/21: [Towards Tuning Tools that "Read" the Manual](#), UC Berkeley (Virtual).
- 2/24/21: SkinnerDB and the Case for Intra-Query Learning, UCSD (Virtual).
- 8/27/20: [SkinnerDB and the Case for Intra-Query Learning](#), Snowflake (Virtual).
- 8/20/20: SkinnerDB and the Case for Intra-Query Learning, Google (Virtual).
- 8/11/20: SkinnerDB and the Case for Intra-Query Learning, Oracle (Virtual).
- 7/31/20: SkinnerDB and the Case for Intra-Query Learning, MSR (Virtual).
- 6/30/20: [Recent Advances in Automated Fact-Checking](#), Global Fact 7 (Virtual).
- 3/5/20: SkinnerDB: the Case for Intra-Query Learning, German Research Foundation, Special Interest Group on Databases.
- 3/3/20: [Exploring Data Using Your Voice](#), NYC Open Data Week.
- 10/14/16: Query Optimization for Data Analysis, UW.
- 4/14/16: Query Optimization for Data Science, UIUC.
- 3/30/16: Query Optimization for Data Science, UCSD.
- 3/22/16: Query Optimization for Data Science, Rice University.
- 3/14/16: Query Optimization for Data Science, ETH Zürich.
- 3/8/16: Query Optimization for Data Science, Cornell.
- 3/3/16: Query Optimization for Data Science, UMD.
- 2/25/16: Query Optimization for Data Science, University of Edinburgh.
- 2/22/16: Query Optimization for Data Science, University of Chicago.

Teaching

At Cornell

- Fall'26: Introduction to Database Systems (CS 4320, 5320). 287 Students Enrolled.
- Fall'26: Practicum in Database Systems (CS 4321, 5321). 50 Students Enrolled.
- Spring'26: Efficient Analysis of Large Data Sets (CS 5322). 31 Students Enrolled.
- Fall'24: Co-taught Introduction to Database Systems (CS 4320, 5320). 417 Students Enrolled.
- Fall'24: Co-taught Practicum in Database Systems (CS 4321, 5321). 49 Students Enrolled.
- Fall'24: Co-taught Seminar in Database Systems (CS 7390). 19 Students Enrolled.
- Spring'24: Advanced Database Systems (CS 6320). 33 Students Enrolled.
- Spring'24: Seminar in Database Systems (CS 7390). 6 Students Enrolled.
- Fall'23: Introduction to Database Systems (CS 4320, 5320). 368 Students Enrolled.
- Fall'23: Practicum in Database Systems (CS 4321, 5321). 25 Students Enrolled.
- Fall'23: Seminar in Database Systems (CS 7390). 7 Students Enrolled.
- Spring'23: Advanced Database Systems (CS 6320). 24 Students Enrolled.
- Spring'23: Seminar in Database Systems (CS 7390). 14 Students Enrolled.
- Fall'22: Introduction to Database Systems (CS 4320, 5320). 346 Students Enrolled.
- Fall'22: Practicum in Database Systems (CS 4321, 5321). 51 Students Enrolled.
- Fall'22: Seminar in Database Systems (CS 7390). 11 Students Enrolled.
- Spring'22: Seminar in Database Systems (CS 7390). 5 Students Enrolled.
- Fall'21: Introduction to Database Systems (CS 4320, 5320). 258 Students Enrolled.
- Fall'21: Seminar in Database Systems (CS 7390): 3 Students Enrolled.
- Spring'21: Advanced Database Systems (CS 6320). 13 Students Enrolled.
- Fall'20: Introduction to Database Systems (CS 4320, 5320). 279 Students Enrolled.
- Fall'20: Practicum in Database Systems (CS 4321, 5321). 33 Students Enrolled.
- Fall'20: Seminar in Database Systems (CS 7390). 7 Students Enrolled.

- Spring'20: Advanced Database Systems (CS 6320). 11 Students Enrolled.
- Fall'19: Introduction to Database Systems (CS 4320, 5320). 276 Students Enrolled.
- Fall'19: Practicum in Database Systems (CS 4321, 5321). 32 Students Enrolled.
- Fall'19: Seminar in Database Systems (CS 7390). 6 Students Enrolled.
- Spring'19: Advanced Database Systems (CS 6320). 14 Students Enrolled.
- Fall'18: Introduction to Database Systems (CS 4320, 5320). 266 Students Enrolled.
- Fall'18: Practicum in Database Systems (CS 4321, 5321). 49 Students Enrolled.
- Fall'18: Seminar in Database Systems (CS 7390). 6 Students Enrolled.
- Spring'18: Advanced Database Systems (CS 6320). 10 Students Enrolled.
- Fall'17: Introduction to Database Systems (CS 4320, 5320). 184 Students Enrolled.
- Fall'17: Practicum in Database Systems (CS 4321, 5321). 26 Students Enrolled.
- Fall'17: Seminar in Database Systems (CS 7390). 5 Students Enrolled.
- Spring'17: Introduction to Database Systems (CS 4320, 5320). 168 Students Enrolled.
- Spring'17: Practicum in Database Systems (CS 4321, 5321). 26 Students Enrolled.
- Fall'16: Advanced Database Systems (CS 6320). 5 Students Enrolled.

External

- [Introduction to Database Systems](#). Online lecture introducing students to topics in database systems with a total duration of 25 hours. The lecture received over **one million** views.
- [The Cornell, Maryland, Max Planck Pre-doctoral Research School 2020](#). I gave a virtual lecture on “Learning to Process Data Faster via Reinforcement Learning”.

Mentoring

PostDocs

- Manuel Schönberger (2026-Present).

Current PhD Students

- Jiale Lao (2024-Present).
- Tharushi Jayasekara (2024-Present).

Graduated PhD Students

- Junxiong Wang (2019-2024). First position: Together AI.
- Saehan Jo (2018-2025). First position: Amazon.
- Victor Giannakouris (2021-2026). First position: Google SRG.
- Ziyun Wei (2018-2024). First position: ByteDance.

Awards by Mentees

- [CRA Outstanding Undergraduate Research Award 2018, Honorable Mention](#). Mark Bryan.
- NSF VLDB 2018 Travel Grant. Samuel Moseley.
- [JP Morgan Award 2018](#). Mark Bryan, Ramya Narasimha.
- [Lockheed Martin Award 2017](#). Mark Bryan, Jiancheng Zhu.

Master of Science Theses

- Ankush Rayabhari. “Adaptive Join Execution in Compilation-Based Execution Engines of Databases”. 2022.

Student Organizations

- [Cornell Data Engineering Team](#), Cornell Data Science (2018-Present).

Master of Engineering Projects

- Ashutosh Agarwal (Spring 2017).
- Niveditha Shenoy Badiadka (Spring 2018).
- Janice Chan (Spring 2020).
- Elva Gao (Fall 2022).
- Shubham Khandelwal (Spring 2019).
- Haram Kim (Fall 2019).
- Daniel Liu (Spring 2017).
- Rahul Madanahalli (Spring 2018).
- Jenny Mallette (Fall 2016).
- Niyati Mehta (Spring 2017).
- Samuel Moseley (Fall 2018).
- Ramya Narasimha (Fall 2017).
- Zane Michael Neelin (Spring 2025).
- Maarij Tahir (Fall 2016).
- Rong Tan (Spring 2020).
- Sanjana Thirumalai (Fall 2016).
- Zhaoyu Wang (Spring 2023).
- Weicheng Yu (Fall 2016).
- Jiancheng Zhu (Fall 2016).

Service

Internal

- Senator-at-Large in the Cornell Faculty Senate (Since 2026).
- CIDA Executive Committee (Since 2025).
- Reviewer for Cisco Strategic Initiative (2023).
- Reviewer for CIDA Research Innovation Fund (2022).
- PhD Admissions Committee (2016, 2017, 2020, 2021, 2022, 2025).
- Recruitment Committee, Ad-Hoc or Full Member (2016-Present).

External

- [Co-Founder and Co-Chair of Q-DATA Workshop \(2024, 2025, 2026\)](#).
- Associate Editor for SIGMOD (2024, 2026).
- Associate Editor for VLDB (2026).
- [Co-Chair of VLDB Demo Track \(2022\)](#).
- [Associate Editor of SIGMOD Record \(2019-Present\)](#).
- Reviewer for IEEE TKDE (2022).
- Panelist for the National Science Foundation (2021, 2023).
- Jim Gray Dissertation Award Committee (2021).
- Reviewer for ACM TODS (2020).
- PC Member for VLDB (2018-Present).
- PC Member for SIGMOD (2018, 2019, 2020, 2026).
- PC Member for SoCC (2017).
- Juror for SIGMOD Undergrad Competition (2016).